

EXERCISE NO.4

1. Write a Java program to create a new array list, add some colors (string) and print out the collection.

Code:

```
import java.util.ArrayList;
import java.util.Iterator;
public class Main {
    public static void main(String[] args) {
        ArrayList<String> list = new ArrayList<>(10);

        list.add("Red");
        list.add("Green");
        list.add("Blue");
        list.add("Yellow");
        Iterator<String> iterator = list.iterator();
        while(iterator.hasNext()){
            String data = iterator.next();
            System.out.println(data);
        }
    }
}
```

2. Modify the above Java program to insert an element into the array list at the first position.

Code:

```
import java.util.ArrayList;
public class Main {
    public static void main(String[] args) {
        ArrayList<String> list = new ArrayList<>(10);
        list.add("Green");
        list.add("Blue");
        list.add("Yellow");
        System.out.println("List Before Inserting Element at 1st Position in ArrayList:"+list);
        list.add(0,"Red");
        System.out.println("List After Inserting Element at 1st Position in ArrayList:"+list);
    }
}
```

3. Modify the above Java program to retrieve an element (at a specified index) from a given array list.

Code:

```
import java.util.ArrayList;
public class Main {
    public static void main(String[] args) {
        ArrayList<String> list = new ArrayList<>();
        list.add("Red");
        list.add("Green");
        list.add("Blue");
        list.add("Yellow");
        String element = list.get(2);
        System.out.println("Retrieve Element is :"+element);
    }
}
```

4. Modify the above Java program to update specific array element by given element.

Code:

```
import java.util.ArrayList;
public class Main {
    public static void main(String[] args) {
        ArrayList<String> list = new ArrayList<>();
        list.add("Red");
        list.add("Green");
        list.add("Blue");
        list.add("Yellow");

        System.out.println("List Before Update:"+list);
        list.set(2, "Black");
        System.out.println("List After Update:"+list);
    }
}
```

5. Modify the above Java program to remove the third element from a array list.

Code:

```
import java.util.ArrayList;
public class Main {
    public static void main(String[] args) {
        ArrayList<String> list = new ArrayList<>();
        list.add("Red");
        list.add("Green");
        list.add("Blue");
        list.add("Yellow");
        System.out.println("Before Removing third element from Array List:"+list);
        list.remove(2);
        System.out.println("After Removing third element from Array List:"+list);
    }
}
```

6. Modify the above Java program to search an element in a array list.

Code:

```
import java.util.ArrayList;
public class Main {
    public static void main(String[] args) {
        ArrayList<String> list = new ArrayList<>();
        list.add("Red");
        list.add("Green");
        list.add("Blue");
        list.add("Yellow");
        if (list.contains("Blue")) {
            System.out.println("Element found");
        } else {
            System.out.println("Element not found");
        }
    }
}
```

7. Modify the above Java program to sort a given array list. (Hint : Use the class Collections).

Code:

```
import java.util.ArrayList;
import java.util.Collections;
public class Program{
    public static void main(String[] args) {
        ArrayList<Integer> list = new ArrayList<>();
        list.add(50);
        list.add(30);
        list.add(20);
        list.add(10);
        list.add(40);
        System.out.println("Before Sorting: " + list);
        Collections.sort(list);
        System.out.println("After Sorting: " + list);
    }
}
```

8. Modify the above Java program to copy one array list into another. (Hint : Use the class Collections).

Code:

```
import java.util.ArrayList;
import java.util.Collections;
public class Program{
    public static void main(String[] args) {
        ArrayList<Integer> source = new ArrayList<>();
        source.add(50);
        source.add(30);
        source.add(20);
        source.add(10);
        source.add(40);

        ArrayList<Integer> destination = new ArrayList<>();
        destination.add(0);
        destination.add(0);
        destination.add(0);
        destination.add(0);
        destination.add(0);
        System.out.println("Source: " + source);
        System.out.println("Before Copy: " + destination);

        Collections.copy(destination,source);
        System.out.println("After Copy: " +destination);
    }
}
```

9. Modify the above Java program to shuffle elements in a array list. (Hint : Use the class Collections).

Code:

```
import java.util.ArrayList;
import java.util.Collections;
public class Program{
    public static void main(String[] args) {
        ArrayList<Integer> list = new ArrayList<>();
        list.add(10);
        list.add(20);
        list.add(30);
        list.add(40);
        list.add(50);

        System.out.println("Before Suffle: " + list);
        Collections.shuffle(list);
        System.out.println("After Suffle: " + list);
    }
}
```

10. Modify the above Java program to reverse elements in a array list. (Hint : Use the class Collections).

Code:

```
import java.util.ArrayList;
import java.util.Collections;
public class Program{
    public static void main(String[] args) {
        ArrayList<Integer> list = new ArrayList<>();
        list.add(10);
        list.add(20);
        list.add(30);
        list.add(40);
        list.add(50);
        System.out.println("Before Reverse: " + list);
        Collections.reverse(list);
        System.out.println("After Reverse: " + list);
    }
}
```

TREESSET

11. Write a Java program to create a new tree set, add some colors (string) and print out the tree set.

Code:

```
import java.util.TreeSet;
public class Program{
    public static void main(String[] args) {
        TreeSet<String> colors = new TreeSet<>();
        colors.add("Red");
        colors.add("Blue");
        colors.add("Green");
        colors.add("Blue");
        colors.add("Yellow");
        System.out.println("TreeSet: " + colors);
    }
}
```

12. Modify the above Java program to add all the elements of a specified tree set to another tree set.

Code:

```
import java.util.TreeSet;
public class Program{
    public static void main(String[] args) {
        TreeSet<String> set1 = new TreeSet<>();
        set1.add("Red");
        set1.add("Blue");
        set1.add("Green");
        TreeSet<String> set2 = new TreeSet<>();
        set2.add("Blue");
        set2.add("Yellow");
        System.out.println("Set 1: " + set1);
        System.out.println("Set 2 Before: " + set2);
        set2.addAll(set1);

        System.out.println("Set 2 After:"+set2);
    }
}
```

13. Modify the above Java program to create a reverse order view of the elements contained in a given tree set.

Code:

```
import java.util.TreeSet;
public class Program{
    public static void main(String[] args) {
        TreeSet<String> colors = new TreeSet<>();
        colors.add("Red");
        colors.add("Blue");
        colors.add("Green");
        colors.add("Blue");
        colors.add("Yellow");

        System.out.println("Original TreeSet : " + colors);
        System.out.println("Reverse Order : " + colors.descendingSet());
    }
}
```

14. Modify the above Java program to get the first and last elements in a tree set.

Code:

```
import java.util.TreeSet;
public class Program{
    public static void main(String[] args) {
        TreeSet<String> colors = new TreeSet<>();
        colors.add("Red");
        colors.add("Blue");
        colors.add("Green");
        colors.add("Blue");
        colors.add("Yellow");
        System.out.println("TreeSet : " + colors);
        System.out.println("First Element: " + colors.first());
        System.out.println("Last Element: " + colors.last());
    }
}
```

15. Write a Java program to get the element in a tree set which is greater than or equal to the given element. (Hint : Use the ceiling method of the TreeSet)

Code:

```
import java.util.TreeSet;
public class Program {
    public static void main(String[] args) {
        TreeSet<Integer> numbers = new TreeSet<>();

        numbers.add(10);
        numbers.add(20);
        numbers.add(30);
        numbers.add(40);
        numbers.add(50);
        int value = 25;

        System.out.println("TreeSet: " + numbers);
        System.out.println("Element greater than or equal to " + value + ": " + numbers.ceiling(value));
    }
}
```